

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 25

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $10^3 / 6^3 = 1000 / 216$ (1) • = 4.6 times (accept 4.63) (1)	2
01.2	<i>1 mark for each point.</i> • C has the highest velocity (most energy, and energy rises with the cube of velocity), strong reliable winds over water, few land-use conflicts, but long expensive cable and offshore installation / maintenance costs (1) • A is windy but on a migration route (bird strikes) and peat bog (foundations / tracks damage hydrology; carbon release); far from grid; scenic objections (1) • B is close to the grid with good wind but near a village (noise, visual objection, shadow flicker) - possible with careful siting (1) • D is cheap to connect and land can still be farmed between turbines, but low velocity gives little energy (1) • Judgement, eg C or B; turbines spaced 3-5 blade diameters apart to avoid wind shadow (1)	5
01.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Driven by wind from any direction so no motor to turn them; suitable for turbulent urban sites (1) • Generator can be at ground level so a lighter tower (1) • Quieter (1) • Operate at lower wind velocities (1) • Blades more visible so fewer bird deaths (1)	2
01.4	<i>1 mark.</i> • No gearbox: fewer breakdowns, less friction so start generating at lower wind speeds, quieter, lighter nacelle (DC converted to AC by an inverter) (1)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 206-209; AQA spec 3.3.4.1.	
02.1	<i>1 mark for each point.</i> - Minimum ozone fell steeply from about 225 DU in 1979 to about 95 DU in the mid 1990s (1) - Since then it has fluctuated around 100-115 DU with a slight recovery / no further decline (1)	2
02.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> $(225 - 95) / 225 \times 100$ (1) $= 57.8\%$ (accept 58%) (1)	2
02.3	<i>1 mark for each point.</i> - Stratospheric temperatures over Antarctica are the lowest on Earth so stratospheric clouds / ice crystals form (1) - Ice crystal surfaces allow reactions that convert the chlorine released from CFCs into chlorine molecules (Cl_2) during the dark winter (1) - The polar vortex (rotating winds) prevents mixing with the rest of the atmosphere, keeping temperatures low and chlorine concentrated (1) - When sunlight returns in spring (September) it splits the chlorine molecules into free radicals that destroy ozone; levels recover by December as the vortex breaks down (1)	4
02.4	<i>1 mark for each point.</i> - Ground and satellite UV / ozone measurements show depletion but not the chemistry / correlation is not causation (1) - Air samples collected from the stratosphere by helium balloons and high-flying research aircraft detected chlorine and chlorine monoxide, confirming the Rowland-Molina hypothesis (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 110-111; AQA spec 3.2.1.3.	
03.1	<i>1 mark for each point.</i> - There is no gaseous / atmospheric reservoir of phosphorus (no equivalent of nitrogen fixation) (1) - Most phosphorus compounds have low solubility so little is available in soil water (1) - Phosphorus lost to marine sediments only returns through very slow processes: mountain building (uplift) and weathering of rocks (1)	3
03.2	<i>1 mark for each point.</i> - Fungal hyphae form a symbiotic relationship with plant roots and spread through a large volume of soil (1) - Greatly increasing the surface area for absorbing (poorly mobile) phosphate ions, passed to the plant in exchange for carbohydrates (1)	2
03.3	<i>1 mark for each point.</i> - Mining of calcium phosphate rock (and past mining of guano deposits, eg Peru, Nauru, now largely exhausted) (1) - Treated to make more soluble ammonium phosphate fertilisers, mobilising phosphorus into the environment (1) - Fertiliser runoff / eroded soil / sewage effluent carry phosphates into water bodies causing eutrophication (with nitrates); harvesting removes phosphorus from farmland (1)	3

Question	Marking guidance	Marks
03.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Use biological wastes (manure, sewage sludge, compost) as fertiliser to return phosphates rather than mining rock (1) • Breed crops that absorb phosphate more efficiently (1) • Maintain conditions for mycorrhizal fungi (less tillage, fewer fungicides, organic matter) (1) • Recover phosphate from sewage effluent by iron(III) sulfate precipitation; reduce soil erosion (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 156-157; AQA spec 3.2.4.4.	
04.1	<i>1 mark for each point.</i> • Readings drift so they may not be accurate; calibration adjusts the meter to match a known standard so results are accurate and comparable between meters / occasions (1) • pH meter placed in buffer solutions of known pH (eg 4, 7, 10) and adjusted to read correctly (1) • If a meter cannot be adjusted, record the percentage error and correct the results (1)	3
04.2	<i>1 mark for each point.</i> • Same height above the ground (1) • No obstacles to air flow; impeller type facing into the wind (1) • Axis of rotation horizontal (impeller) or vertical (cup); same time period / averaging (1)	3
04.3	<i>1 mark for each point.</i> • Two thermometers, one with a wet bulb, are spun until the wet bulb stops cooling (1) • Evaporation cools the wet bulb more when humidity is low; the temperature difference is compared with a table to find humidity (1)	2
04.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Test strips with reagent bands compared with a colour chart (semi-quantitative) (1) • Electronic colourimeter: reagent added and colour intensity measured at a specific wavelength (1) • Ion-selective electrode measuring electrode potential (other ions can interfere) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 407-410, 413; AQA spec 3.7.2.1.	
05.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(15.0 - 2.2) / 15.0 \times 100$ (1) • = 85.3% (accept 85%) (1)	2
05.2	<i>1 mark for each point.</i> • Added as an anti-knock agent to smooth combustion and reduce engine wear (1) • Lead particulates were released in exhaust gases into the air of cities and inhaled / deposited on soil and food (1) • Lead is a neurotoxin that bioaccumulates; chronic exposure damages the nervous system / brain development in children (1)	3

Question	Marking guidance	Marks
05.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Replacement of lead water pipes / lead solder with copper / tin-based solder; phosphoric acid dosing of water supplies forming insoluble lead phosphate (1) • Phasing out of lead-based paint / safer paint removal (1) • Lead-free electrical solder (1) • Restrictions on lead fishing weights / shotgun pellets (1) • Industrial emission controls (1)	2
05.4	<i>1 mark for each point.</i> • Reacts with the lead in old pipes to form an insoluble layer of lead phosphate on the inside of the pipe (1) • This prevents lead dissolving into the water (1)	2
05.5	<i>1 mark.</i> • Swans / ducks swallow them (mistaking them for snails / grit) and the lead is absorbed, causing lead poisoning (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-276; AQA spec 3.4.3.2.11.	
06.1	<i>Correct answer only.</i> • C (1)	1
06.2	<i>1 mark for each point.</i> • Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate (1) • No need to sample sites that were never likely to be polluted; sampling also checks the accuracy of the predictions (1)	2
06.3	<i>1 mark for each point.</i> • Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify (1) • So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	
07.1	<i>1 mark for each point.</i> • Natural pressure of gas above the oil (or dissolved in it) or water beneath forces oil up the production well; pump-jacks ('nodding donkeys') increase flow (1) • Pressure falls as oil is removed so the remaining oil no longer flows to the well (1)	2
07.2	<i>1 mark for each point.</i> • Water or natural gas is pumped down an injection well to maintain the pressure and flow of oil (recovery to about 40%) (1) • Captured CO₂ can be pumped down instead, increasing oil recovery while storing the CO₂ underground (1)	2
07.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Steam injection (from combustion or solar parabolic concentrators) to heat the oil and reduce its viscosity (1) • Controlled underground combustion to heat the oil (1) • Detergents / solvents to reduce surface tension so oil flows more easily (1) • Bacteria that partially digest heavy oil into lighter oils and produce CO₂ to maintain pressure (1)	3

Question	Marking guidance	Marks
07.4	<i>1 mark for each point.</i> • Many wells can be drilled from a single platform, reducing land take / number of rigs (1) • Deposits under locations where rigs cannot be placed (urban areas, sensitive habitats) can be reached (1) • Drilling can follow softer strata (quicker) and target multiple small reservoirs up to 10 km away, increasing total recovery (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187-188; AQA spec 3.3.4.1.	
08.1	<i>Correct answer only.</i> • B - rivers are accessible over a large area with few natural contaminants (1)	1
08.2	<i>1 mark for each point.</i> • Potable / public supply: no pathogens, low toxins, good appearance / taste / smell (public health) (1) • Industrial boiler / cooling water: soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange; spray irrigation: low turbidity to avoid blocked pipes and low heavy metals (1)	2
08.3	<i>1 mark for each point.</i> • Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) (1) • Disadvantage: high cost / energy for pumping; ecological impacts on the donor river; transfer of pollutants or invasive species; political conflict (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
09.1	<i>1 mark for each point.</i> • Surplus electricity (eg windy nights) electrolyses water to produce hydrogen which is stored (1) • Hydrogen has a high energy density so it can be transported / stored for long periods (1) • Released by combustion for heat, industrial processes or steam turbines; or by fuel cells combining hydrogen and oxygen electrochemically to produce electricity and water (1) • Can also be used as a vehicle fuel or converted to methane / ammonia (1)	4
09.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Compressed gas (up to 700 times normal volume) - energy to run the compressor, strong tanks (1) • Liquefied at very low temperature - energy for refrigeration and compression (1) • Adsorbed on a metal matrix (metal hydride) at low pressure - tanks larger and heavier than a petrol tank storing the same energy (1) • Conversion to ammonia - some energy needed but less than the alternatives (1)	3
09.3	<i>1 mark for each point.</i> • Both convert chemical energy to electricity electrochemically, but a battery stores a fixed amount of chemicals which are reformed by recharging (1) • A fuel cell keeps producing electricity as long as fresh fuel (hydrogen / alcohol) and oxidant are supplied (1)	2

Question	Marking guidance	Marks
09.4	<i>1 mark.</i> Using hydrogen made from surplus renewable electricity as the main store and carrier of energy, released when demand exceeds supply or intermittent sources fail, for heating, vehicles and electricity (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 219-221; AQA spec 3.3.4.1.	
10.1	<i>1 mark for each point.</i> - Together they form a dispersed layer in the stratosphere (the ozone layer / ozonosphere) (1) - They absorb ultraviolet light, which drives a dynamic equilibrium of reactions that form and destroy ozone (1) - The UV energy is converted to chemical / heat energy so most UV that passed the upper atmosphere never reaches the surface (1)	3
10.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Provides carbon, oxygen, hydrogen and nitrogen (as N₂, O₂, CO₂, H₂O) for biological molecules (1) - Delays the escape of infrared energy, keeping the surface warm (1) - Distributes heat and water vapour by winds / ocean currents (1) - Provides atmospheric pressure so liquid water can exist (1) - Deflects damaging radiation of the solar wind in the upper atmosphere (1)	2
10.3	<i>1 mark.</i> Energy is absorbed at different altitudes (eg UV absorbed in the stratosphere, surface heating warms the troposphere) so temperature gradients change with altitude, producing layers (1)	1

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Troposphere: enhanced greenhouse effect - anthropogenic CO₂ (fossil fuels, deforestation), methane (rice, livestock, landfill), N₂O, CFCs, tropospheric ozone; more infrared absorbed and re-emitted; aerosols and smoke increase albedo / scatter light; land-use change alters surface albedo and evapotranspiration • Stratosphere: CFCs / halons / HCFCs release chlorine and bromine which catalyse ozone destruction, shifting the O / O₂ / O₃ equilibrium; less UV absorbed in the stratosphere, more reaches the surface; stratospheric cooling • Consequences: changes to climate, ocean currents (thermohaline), hydrological cycle, sea level, species distributions; increased UV damage to organisms • Difficulties of prediction: linked processes where one change alters others; feedback (positive and negative) and tipping points; time lags (ocean heat capacity, long residence times of CFCs and CO₂); natural variability (solar output, volcanoes, Milankovitch cycles); incomplete understanding of atmospheric processes; limited historical data so proxies needed; models with different assumptions • Evaluation: some effects well understood (greenhouse physics, ozone chemistry proven, Montreal Protocol success) but scale and timing of regional impacts uncertain; precautionary principle • Justified conclusion	9
	Model answer - Level 3 (8 marks)	
	In the troposphere human activity enhances the greenhouse effect: fossil fuels and deforestation add CO ₂ , rice, livestock and landfill add methane, fertilisers add nitrous oxide, and CFCs and tropospheric ozone are new absorbers. More outgoing infrared is absorbed and re-emitted, warming the surface, while smoke and aerosols scatter sunlight and land-use change alters albedo and evapotranspiration. In the stratosphere, CFCs, halons and HCFCs release chlorine and bromine that catalyse ozone destruction, shifting the O / O ₂ / O ₃ equilibrium so that less UV is absorbed there and more reaches the surface, cooling the stratosphere as well.	

Question	Marking guidance	Marks
	The consequences run through every system the atmosphere's energy drives: climate, thermohaline circulation, the hydrological cycle, sea level and species distributions, plus UV damage to organisms. Predicting them is hard because the processes are linked - one change alters others - and because feedback can amplify (albedo loss, methane release) or damp (cloud cover, photosynthesis) the effect, sometimes past tipping points. Time lags from ocean heat capacity and the long residence times of CO ₂ and CFCs, natural variability from solar output, volcanoes and orbital cycles, sparse historical data replaced by proxies, and models built on different assumptions all add uncertainty.	
	Some things are nevertheless well established: the physics of infrared absorption, the chemistry of ozone loss, and the success of the Montreal Protocol. What is uncertain is the scale and timing of regional impacts, which is exactly why the precautionary principle, not certainty, should guide action.	
	Weaker answer - Level 1 (3 marks)	
	Humans release greenhouse gases like carbon dioxide from burning fossil fuels into the troposphere, which traps more heat and causes global warming. CFCs go into the stratosphere and destroy the ozone layer, so more UV reaches the Earth. It is hard to predict what will happen because the atmosphere is complicated and scientists do not know everything about it. There could be feedback effects that make it worse.	
	What makes the difference	
	• Level 1 gives one cause per layer; Level 3 lists several gases and processes and explains the mechanism (absorption and re-emission; catalytic destruction shifting an equilibrium).	
	• 'Difficulties of prediction' is answered in Level 3 with specific reasons (linked processes, feedback, time lags, natural variability, proxy data, models); Level 1 says 'complicated'.	
	• Level 3 evaluates - separating what is certain from what is not - and links this to the precautionary principle.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89, 108-113; AQA spec 3.2.1.1, 3.2.1.3.	

Question	Marking guidance	Marks
11.1	Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Attributes: accuracy, precision, representative samples, reliability, anomalies, validity; hypothesis and null hypothesis; preliminary studies • Sampling design: random vs systematic location and avoiding bias; timing (diurnal, seasonal, long-term, weather); sample size and number and the running mean; standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • Statistics: variability, standard deviation and error bars, significance and p-values, correlation vs causation; choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • Application to Paper 1 topics: monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	25

Question	Marking guidance	Marks
11.2	Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Hydrological cycle processes and reservoirs; residence times; dynamic equilibrium; increasing demand (population, affluence, irrigation, industry) • Human impacts on transfer rates: deforestation, agriculture, urbanisation, climate change, abstraction; consequences for rivers, aquifers (salt incursion, subsidence, wetland loss), reservoirs • Case studies of unsustainable exploitation: Aral Sea, Nile, Tibetan rivers, Ogallala, North China Plain, Malta • Sustainable strategies: metering, low-water appliances, greywater, xeriscaping, drip irrigation, leak repair; new sources: rainwater catchment, reservoirs and estuary barrages (site factors), unexploited aquifers, inter-basin transfers, artificial recharge, river regulation, afforestation, desalination • Water treatment to make more water usable: sedimentation, flocculation, filtration, activated carbon, sterilisation, ion exchange, reverse osmosis; pollution control (sewage, AMD, buffer strips) • Links to thermohaline circulation and climate; energy costs; circular economy principles (catchment management) • Evaluation of which strategies are most sustainable in different regions; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-128; AQA spec 3.2.2.	25